

CPU Scheduling Simulator

Test Cases Document

Priority Scheduling vs SRTF | OS Project C1

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Project Title	CPU Scheduling Simulator
Algorithms	Preemptive Priority Scheduling vs SRTF
Repository	github.com/abdallah-wael-1/os-project

Overview

This document presents the four test scenarios used to validate the CPU Scheduling Simulator. Each scenario is described with its purpose, input processes, expected behaviour for both Preemptive Priority Scheduling and SRTF, and the expected output metrics. Screenshots for each scenario are submitted separately.

Scenario A — Basic Mixed Workload

Purpose

This scenario tests the fundamental behaviour of both algorithms with a standard set of processes that have different arrival times, burst times, and priorities. It verifies that each algorithm correctly selects the next process to run at every time unit, handles preemption on new arrivals, and produces accurate completion, turnaround, waiting, and response time values.

Input

PID	Arrival Time	Burst Time	Priority
P1	0	8	3
P2	1	4	1
P3	2	9	4
P4	3	5	2

Expected Behaviour — Priority Scheduling

P1 starts running at t=0 since it is the only process in the ready queue. At t=1, P2 arrives with priority 1 (the highest possible), so it immediately preempts P1. P2 runs from t=1 to t=5 and completes. At t=3, P4 arrives with priority 2, and at t=5 when the CPU becomes free, P4 is selected over P1 (priority 3) and P3

(priority 4). P4 runs from $t=5$ to $t=10$ and completes. P1 then resumes from where it was preempted and runs from $t=10$ to $t=17$. Finally, P3 runs from $t=17$ to $t=26$.

Expected Behaviour — SRTF

P1 starts at $t=0$ with burst 8. At $t=1$, P2 arrives with burst 4, which is shorter than P1's remaining burst of 7, so P2 preempts P1. At $t=2$, P3 arrives with burst 9 — longer than P2's remaining 3, so no preemption. At $t=3$, P4 arrives with burst 5, also longer than P2's remaining 2, so P2 continues. P2 completes at $t=5$. At $t=5$, the ready queue contains P1 (remaining 7), P3 (9), and P4 (5). P4 has the shortest remaining time so it runs from $t=5$ to $t=10$. P1 (remaining 7) is next, running from $t=10$ to $t=17$. Finally P3 runs from $t=17$ to $t=26$.

Expected Output Metrics

Both algorithms produce the same metrics for this scenario:

Process	CT	TAT	WT	RT
P1	17	17	9	0
P2	5	4	0	0
P3	26	24	15	15
P4	10	7	2	2
Average	—	13.00	6.50	4.25

Scenario B — Priority vs Burst Conflict

Purpose

This scenario is designed to reveal the core difference between the two algorithms by creating a direct conflict: the highest-priority process has the longest burst time, while the lowest-priority processes have very short burst times. The goal is to show that Priority Scheduling and SRTF make completely opposite decisions in this situation, resulting in different execution orders and different performance metrics.

Input

PID	Arrival Time	Burst Time	Priority
P1	0	15	1 (highest priority)
P2	1	2	5 (lowest priority)
P3	2	3	5 (lowest priority)

Expected Behaviour — Priority Scheduling

P1 starts at $t=0$ and holds priority 1, the highest priority in the system. When P2 arrives at $t=1$ with priority 5, it cannot preempt P1 because its priority number is higher (lower urgency). When P3 arrives at $t=2$ with priority 5, the same rule applies. P1 runs uninterrupted from $t=0$ to $t=15$ and completes. The CPU then picks P2 (priority 5, arrived earlier) and runs it from $t=15$ to $t=17$. Finally P3 runs from $t=17$ to $t=20$.

Expected Behaviour — SRTF

P1 starts at $t=0$. At $t=1$, P2 arrives with burst 2, which is much shorter than P1's remaining burst of 14. SRTF ignores priority entirely and preempts P1 immediately. P2 runs from $t=1$ to $t=3$ and completes. At $t=2$, P3 had arrived with burst 3. At $t=3$, the ready queue has P1 (remaining 14) and P3 (remaining 3). P3 is shorter, so it runs from $t=3$ to $t=6$. P1 then resumes with remaining burst 14 and runs from $t=6$ to $t=20$.

Expected Output Metrics

Priority Scheduling:

Process	CT	TAT	WT	RT
P1	15	15	0	0
P2	17	16	14	14
P3	20	18	15	15
Average	—	16.33	9.67	9.67

SRTF:

Process	CT	TAT	WT	RT
P1	20	20	5	0
P2	3	2	0	0
P3	6	4	1	1
Average	—	8.67	2.00	0.33

Scenario C — Starvation Risk

Purpose

This scenario demonstrates the starvation problem that can arise in Preemptive Priority Scheduling. All four processes arrive at the same time ($t=0$), so the CPU immediately selects based on priority. The process with the lowest priority (P4, priority 5) must wait the longest. The scenario also shows how SRTF handles the same workload differently by scheduling based on burst length rather than priority, which distributes CPU time more fairly among short processes.

Input

PID	Arrival Time	Burst Time	Priority
P1	0	10	1
P2	0	5	2
P3	0	3	3
P4	0	8	5 (starvation risk)

Expected Behaviour — Priority Scheduling

All processes arrive at $t=0$. The scheduler selects P1 (priority 1) first. Since no new arrivals occur, P1 runs uninterrupted from $t=0$ to $t=10$. Next, P2 (priority 2) runs from $t=10$ to $t=15$. Then P3 (priority 3) runs from $t=15$ to $t=18$. Finally P4 (priority 5) runs from $t=18$ to $t=26$. P4 had to wait 18 time units before getting any CPU time. If higher-priority processes kept arriving indefinitely, P4 would never run — this is the starvation problem.

Expected Behaviour — SRTF

All processes arrive at $t=0$. SRTF selects P3 first (shortest burst = 3). P3 runs from $t=0$ to $t=3$ and completes. Next, P2 (burst 5) runs from $t=3$ to $t=8$. Then P4 (burst 8) runs from $t=8$ to $t=16$. Finally P1 (burst 10) runs from $t=16$ to $t=26$. No process is starved — every process runs as soon as all shorter ones complete.

Expected Output Metrics

Priority Scheduling:

Process	CT	TAT	WT	RT
P1	10	10	0	0
P2	15	15	10	10
P3	18	18	15	15
P4	26	26	18	18
Average	—	17.25	10.75	10.75

SRTF:

Process	CT	TAT	WT	RT
P1	26	26	16	16
P2	8	8	3	3
P3	3	3	0	0
P4	16	16	8	8
Average	—	13.25	6.75	6.75

Scenario D — Input Validation

Purpose

This scenario verifies that the simulator correctly rejects all forms of invalid input before allowing a simulation to run. Seven distinct error cases are tested. Each case is triggered through dedicated buttons in the Scenario D panel of the simulator. The expected result for every case is a descriptive inline error message with a shake animation, and the simulation must not proceed until the input is corrected.

Validation Test Cases

#	Invalid Input	Input Used	Expected Error
1	Negative arrival time	Arrival = -1	Arrival time must be ≥ 0
2	Burst time = 0	Burst = 0	Burst time must be > 0
3	Duplicate Process ID	Two processes with same PID	Process ID must be unique
4	Non-numeric value	Burst = "abc"	Please enter a valid number
5	Missing required fields	Empty arrival/burst field	All fields are required
6	Priority out of range	Priority = 0 or 100	Priority must be between 1 and 99
7	Fewer than 2 processes	Only 1 process added	At least 2 processes are required

Expected Behaviour

For each error case, the simulator must display an inline error message next to the relevant field with a shake animation. The Run Simulation button must remain disabled or produce no output until all validation errors are resolved. No partial simulation results should appear when invalid input is present.